

ECE 501 DIGITAL IMAGE PROCESSING
SAR Optical Feature Fusion For Crop Type Classification
REPORT WEEK 3
GROUP 11

Ahladita Chaudhary	AU2440243
Priyanshi Joshi	AU2440092
Naisha Dave	AU2440181
Margi Trivedi	AU2440035

1. What is the crop-classification pipeline?

The crop-classification pipeline involves preparing the data, extracting features from satellite images, combining the features, and using them to classify crops. In our project, we are using Sentinel-2 optical data and Sentinel-1 SAR data to classify corn, soybean, and wheat.

The workflow is: YieldSAT data -> Data preprocessing -> NDVI and VV/VH extraction -> Feature fusion -> Crop classification.

2. What is NDVI, and how is it calculated?

NDVI (Normalized Difference Vegetation Index) is used to study vegetation using red and near-infrared reflectance from satellite images.

The formula is: $NDVI = (B08 - B04) / (B08 + B04)$

Here, B08 is the near-infrared band and B04 is the red band of Sentinel-2.

We calculated and verified NDVI for 100 optical records from Argentina and Brazil. The dataset contains 34 corn, 34 soybean, and 32 wheat samples. The NDVI values ranged from 0.188 to 0.676, with a mean of 0.357. We also created a histogram and a crop-wise boxplot to study the distribution of NDVI values across the three crop classes.

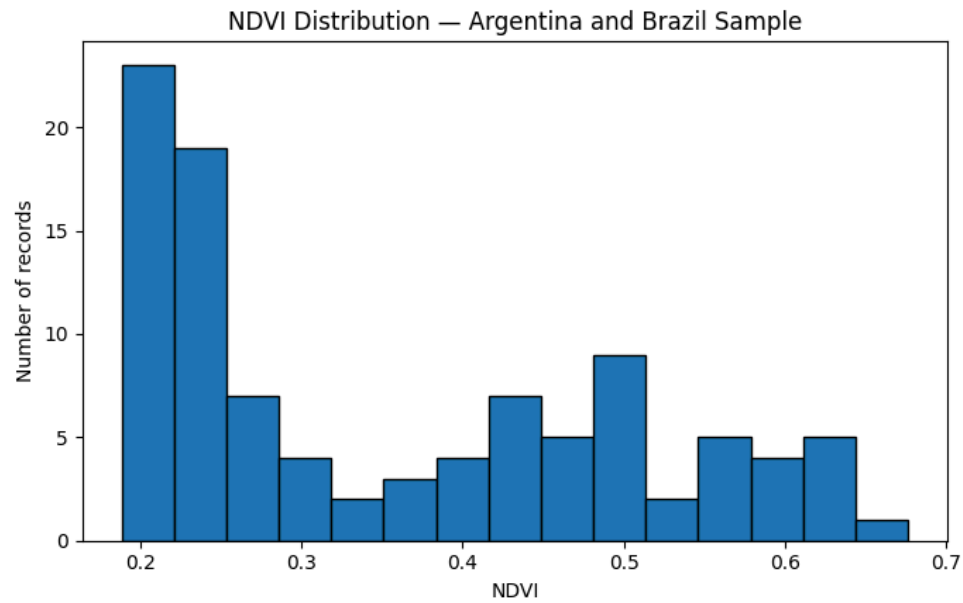


Figure 1: Distribution of NDVI values across the 100 optical samples

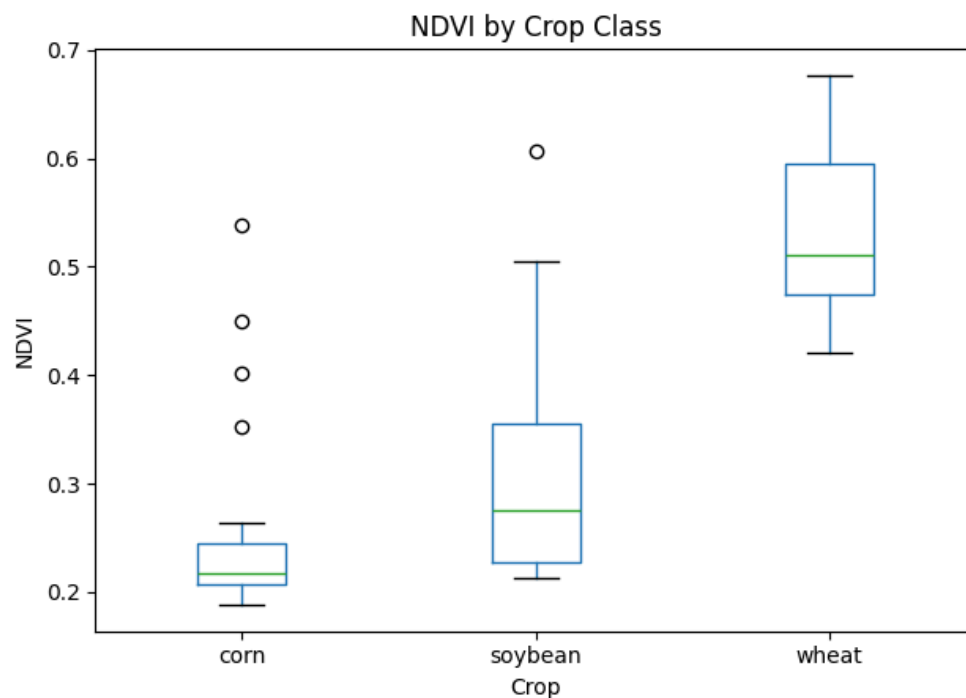


Figure 2: Comparison of NDVI distributions for corn, soybean, and wheat

3. How will crop labels be linked with extracted features?

Each sample will be identified using fields such as index and field_shared_name, while the crop column will provide its crop label.

The NDVI values will be linked to their respective samples. Similarly, Sentinel-1 VV and VH values will be extracted for the same locations and matched using sample IDs, verified coordinates, and compatible dates. These features will then be combined into one table containing the sample ID, crop label, NDVI, VV, and VH. This table will be used for crop classification.

4. How were Argentina and Brazil datasets extracted into CSV files?

The Argentina and Brazil YieldSAT datasets have been processed from their respective NetCDF files using Python in Google Colab. The required records were extracted and stored into separate country-sensible CSV documents named `Argentina_with_coordinates.csv` and `Brazil_with_coordinates.csv` for in addition analysis.

5. How were coordinates extracted and validated?

The coordinate additives `coord_x`, `coord_y`, and `coord_z` were extracted and blanketed inside the countrywise csv files. Validation assessments had been carried out to pick out lacking coordinate components. The processing logs reported 0 lacking values for all 3 coordinate components in both countries.

6. What challenges were faced during data processing?

The important mission is managing the massive NetCDF files whilst working with restrained storage and computational sources. Google Colab and a country-wise processing method have been used to control the records extraction more successfully and prepare the datasets for next evaluation.

7. How is Sentinel-1 data accessed using Google Earth Engine?

Sentinel-1 Synthetic Aperture Radar (SAR) data were accessed using Google Earth Engine (GEE), in the `COPERNICUS/S1_GRD` image collection. The collection offers calibrated radar backscatter imagery that can be analyzed with remote sensing. An end date of 1 October 2016 was used and applied as an exclusive end date in the date filter for September 2016 (1 September - 30 September). In addition, the data was further subset to only include acquisitions obtained in Interferometric Wide (IW) mode and scenes with both VV and VH polarization bands. The collection was sorted by the acquisition date and the first matching image was

chosen for the initial inspection. This Sentinel-1A scene was acquired on 1st September 2016. This number of scenes in the filtered collection did not reflect the number of scenes covering the study fields, it was the total number of scenes worldwide. GEE was also used to show the radar bands and to set up an export for GeoTIFF for additional analysis.

8. How are VV and VH polarization bands retrieved?

The VV and VH polarization bands were obtained by applying a filter to the Sentinel-1 collection to keep all scenes with both polarizations and selecting the two bands with `.select(['VV', 'VH'])`.

VV (Vertical transmit, Vertical receive): Vertical transmit and receive (Radar backscatter measured with the vertical transmit and receive).

VH (Vertical transmit, Horizontal receive): Vertical transmit, horizontal receive, radar backscatter measured in this way.

The image chosen was checked to ensure that it was acquired on the correct date and the correct bands. A map layer for VV and a map layer for VH were created. The ranges of the visualization were -25 to 0 dB for VV, and -30 to -5 dB for VH.

The Sentinel-1 GRD VV and VH values are expressed as calibrated backscatter in decibels (dB). No further scaling was used for the script in the current project. A test GeoTIFF export was set up at a specified scale of 10 metres, over a small area around the centroid of the image footprint selected. This was performed as a technical test export and not as a full raster of the Brazil and Argentina sample fields.

9. How will SAR data be used in crop classification?

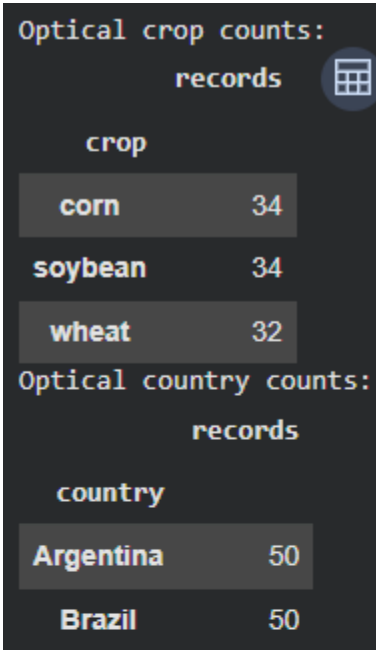
SAR data may be used to assist in the classification of crop type as it reflects the structure of vegetation, the canopy properties, the surface roughness, and moisture in the vegetation. VV and VH therefore can give additional information in addition to optical characteristics like NDVI.

The intended procedure will be to extract VV-VH values at the same valid sample locations as used in the optical workflow, assign these features to the appropriate sample position ID and crop labels, and then merge these

into a feature table to be used for classification. A temporal SAR based feature or composite can be used if the acquisition date and data availability is appropriate. But at present, the extraction from the field is not valid VV and VH values for the uploaded sample geometries. The first inspected Brazil and Argentina had empty geometries, and the first field level extraction yielded no valid VV and VH data. Thus, the real case of the fusion of SAR and optical data at the field level has yet to be realized. This is possible only after verifying the validity of the sample coordinates and the corresponding identifiers.

10. How were Sentinel-1 and Sentinel-2 data inspected?

For the optical data, we loaded the Argentina and Brazil datasets and checked the records, crop labels, missing values, and duplicates. The dataset had 100 records in total, including 34 corn, 34 soybean, and 32 wheat records.



The screenshot shows a dark-themed interface with two sections. The first section, titled 'Optical crop counts:', has a sub-header 'records' and a table with three rows: 'corn' with 34 records, 'soybean' with 34 records, and 'wheat' with 32 records. The second section, titled 'Optical country counts:', has a sub-header 'records' and a table with two rows: 'Argentina' with 50 records and 'Brazil' with 50 records. A small grid icon is visible in the top right corner of the first section.

Optical crop counts:	
records	
crop	
corn	34
soybean	34
wheat	32

Optical country counts:	
records	
country	
Argentina	50
Brazil	50

We calculated NDVI using the red band (B04) and near-infrared band (B08) with the formula:

$$\text{NDVI} = (\text{B08} - \text{B04}) / (\text{B08} + \text{B04})$$

We saved the processed data in `Optical_Features.csv` and created graphs to study the NDVI distribution, compare NDVI across crops, and look at the band values.

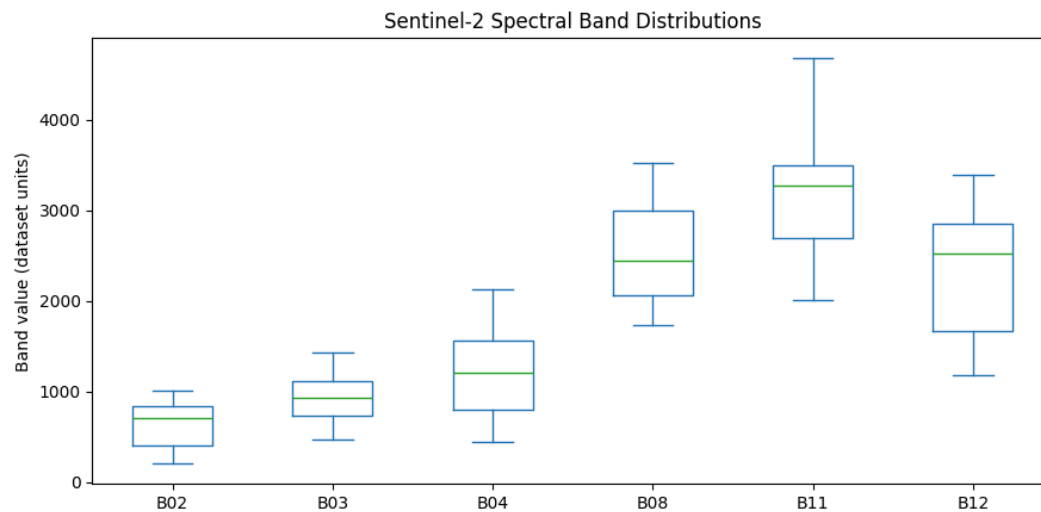


Figure 3: Distribution of Sentinel-2 spectral band values across the optical samples

For the SAR data, we checked the available test raster and its VV and VH bands. It was an 11 x 11 pixel raster with a pixel size of 10 metres. The average VV value was around -13.91 dB, while VH was around -20.08 dB. These values are from the test raster and have not been linked to our actual field records.

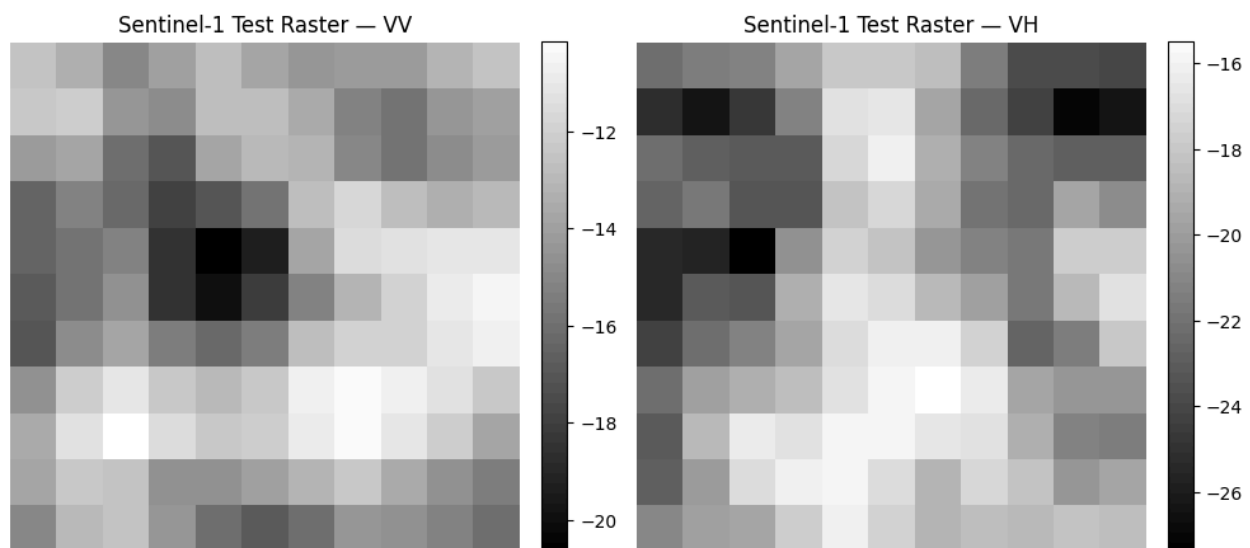


Figure 4: VV and VH backscatter visualization of the Sentinel-1 test raster.

11. What data-quality or alignment issues were identified?

During the checks, we noticed a few things that need attention before we can combine the datasets. The latest SAR CSV has empty VV and VH columns, and the geometry information appears to be missing. Some records also have differences between their date and year values. The optical and SAR data also need to be matched using the correct field IDs, locations, and dates before we can use them together.

12. What preprocessing is required before feature extraction?

Before extracting and combining the features, a few preprocessing steps are needed to make sure the data is accurate and consistent. First, we need to check that the field IDs, crop labels, dates, and years are correct. We also need valid field coordinates or boundaries so that the satellite data can be matched to the right fields.

Next, the optical data should be checked for missing values, and we need to confirm the band values and their scaling before using them. The NDVI calculations should also be verified, and any invalid or missing pixels should be handled properly.

For the SAR data, we need to obtain the actual VV and VH values for our fields and check the raster's coordinate system, resolution, and missing-data handling. Once both datasets are ready, we can match the Sentinel-1 and Sentinel-2 observations using the correct field IDs, locations, and acquisition dates. Finally, we can combine the NDVI, VV, and VH values with the field IDs and crop labels into a single table, while clearly identifying any missing or unmatched data.

13. Work Done this Week

During Week 3, the group started out the initial implementation of the crop-class pipeline. We labored on getting ready the Argentina and Brazil YieldSAT datasets and producing countrywise csv files with coordinate statistics. The coordinate components have been checked for missing values, with zero missing components suggested in the processing logs. We focused on the initial pipeline setup and optical feature extraction, We also labored on Sentinel-1 SAR information acquisition using Google Earth Engine, and then targeted on satellite tv for pc information inspection and preprocessing. These obligations help the development of a commonplace dataset for comparing optical-simplest, SAR-simplest, and blended SAR-optical methods.

14. Work Planned for Next Week

The foremost intention for Week 4 is to finish the preliminary function-extraction pipeline. For optical records, the organization will put into effect NDVI, NDRE, NDMI, EVI, basic statistical features, and GLCM texture functions. For SAR information, the group will extract VV, VH, the VV/VH ratio, statistical features, and GLCM texture capabilities. The institution can even put into effect missing-cost dealing with, invalid-sample filtering, regular subject IDs and function names, and merging of extracted functions with crop labels. The whole workflow will first be examined on a small subset of the dataset. After validating the consequences, the pipeline may be scaled to the whole usable dataset.

15. Challenges Faced

During this phase, we worked on understanding and preparing the optical and SAR data for further analysis. We were able to calculate NDVI from the optical data and create visualizations to study the results. We also explored the available SAR data to understand its structure and values. The main challenge was figuring out how to bring both datasets together, since the field information, locations, and acquisition dates need to match. Making sure the two datasets are properly aligned will be the next important step before using the combined features for crop classification.

16. Conclusion

In Week 3, we worked on the optical data by calculating NDVI and making graphs to understand the results. we also checked the available SAR data and looked at the VV and VH values. This helped me get a better understanding of both datasets. The next step is to match the SAR data with the correct field records and prepare a combined dataset with the optical and SAR features. This will give us a better starting point for moving towards crop classification.